



Amazon Bedrock Getting Started

Introduction

- Amazon Bedrock is a fully managed service that offers leading foundation models (FMs) and a set of capabilities to quickly build and scale generative artificial intelligence (generative AI) applications.
- The service also helps ensure privacy and security.
- Choose from various FMs to find the model that is best suited for your use case.
- With the Amazon Bedrock serverless experience, you can quickly get started and use FMs without the need to manage any infrastructure.
- You can also privately customize FMs with your own data and seamlessly integrate and deploy them into your applications by using Amazon Web Services (AWS) tools and capabilities.

Capabilities of Amazon Bedrock

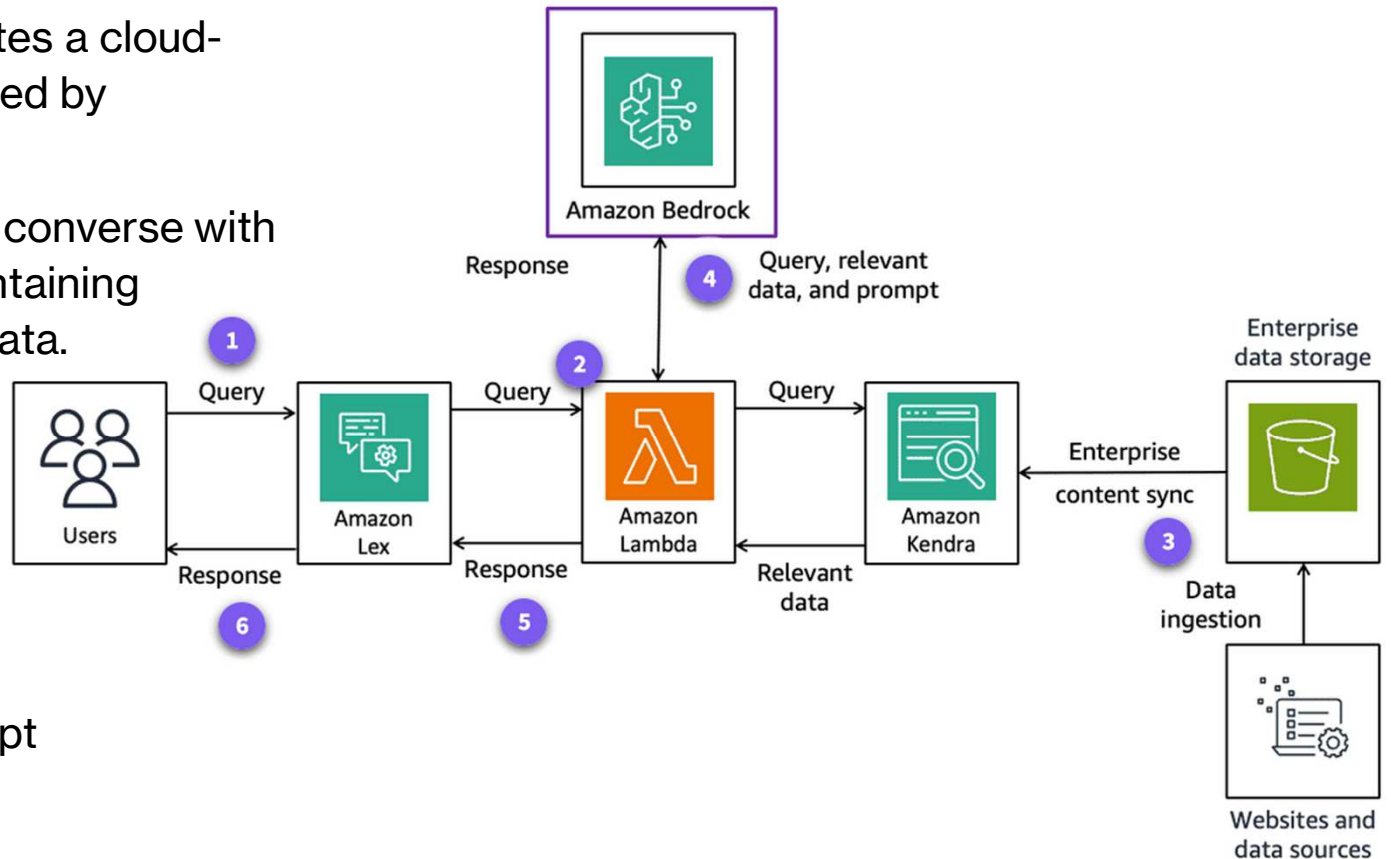
- **Foundation models** that include a choice of base FMs and customized FMs
- **Playgrounds** for chat, text, and images with quick access to FMs for experimentation and use through the console
- **Safeguards** such as watermark detection and guardrails
- **Orchestration and automation** for your application with knowledge bases and agents
- **Assessment and deployment** with model evaluation and provisioned throughput

Architecture and Use Cases

The following architecture illustrates a cloud-based AI assistant solution powered by generative AI and customer data.

This solution has the capability to converse with you in natural language while maintaining accuracy by including customer data.

1. User Query
2. Compute and business logic
3. Relevant data
4. Query, relevant data and prompt
5. Response
6. Response



What are typical use cases for Amazon Bedrock?



Text generation

Create new pieces of original content, such as short stories, essays, social media posts, and webpage copy.



Virtual assistants

Build assistants that understand user requests, automatically break down tasks, engage in dialogue to collect information, and take actions to fulfill requests.



Text and image search

Search and synthesize relevant information to answer questions and provide recommendations from a large amount of text and image data.



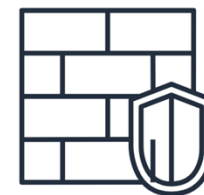
Text summarization

Get concise summaries of long documents, such as articles, reports, research papers, technical documentation, and books, to quickly and effectively extract important information.



Image generation

Quickly create realistic and visually appealing images for advertising campaigns, websites, presentations, and more.



Guardrails

Implement safeguards customized to your application requirements and responsible artificial intelligence (AI) policies.

How much does Amazon Bedrock cost?

On-Demand Mode

- You are billed based on usage: per input/output token for text models, per input token for embeddings models and per image for image generation.
- There are no time-based commitments – costs apply only to actual usage, offering flexibility and scalability.

Provisioned Throughput Mode Summary:

- This mode allows you to purchase model units for base or custom models, offering guaranteed throughput for large, consistent inference workloads.
- Throughput is measured in tokens per minute and custom models are accessible exclusively through this mode.

How Do You Use Amazon Bedrock?

You can use Amazon Bedrock from the AWS Management Console or with an API.

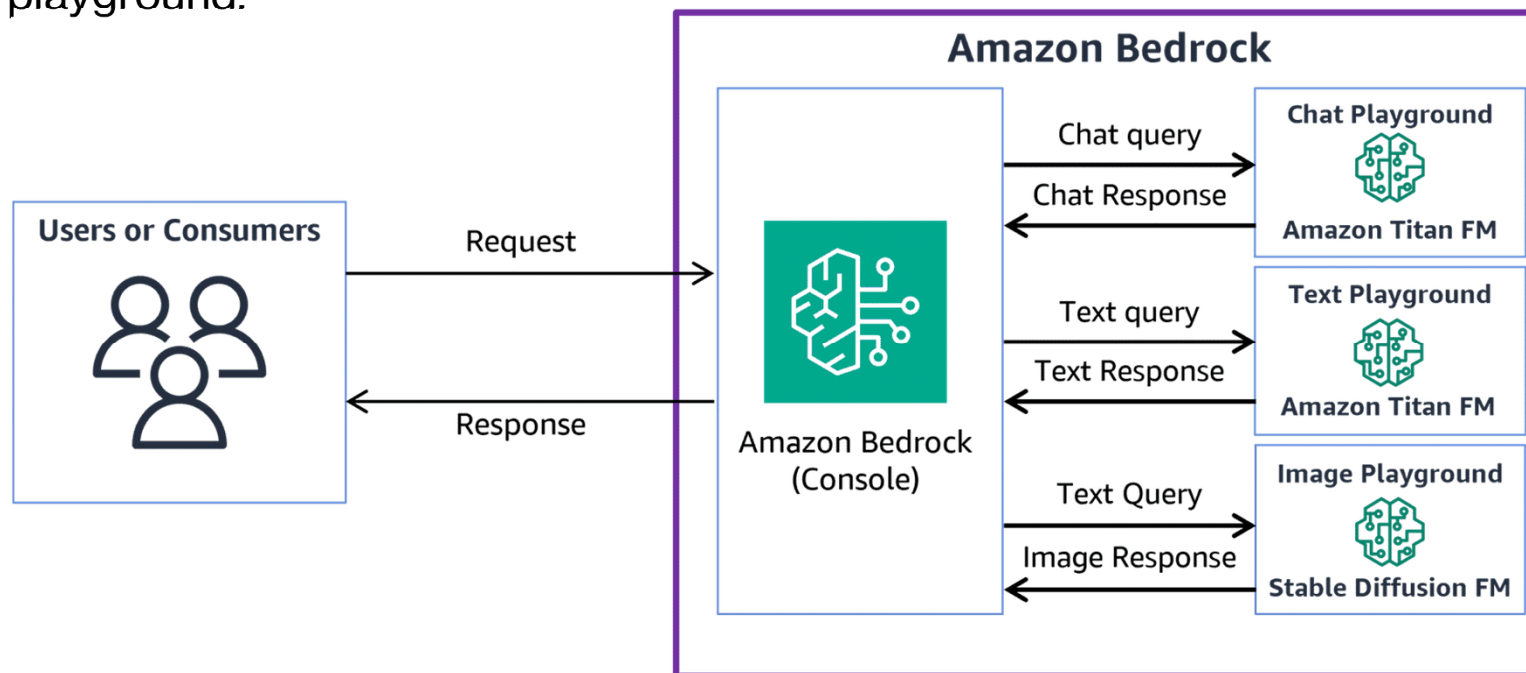
Amazon Bedrock Amazon Bedrock playgrounds let you interact with foundation models (FMs) to generate text, images, summaries, answers, or engage in chat, with support for multiple model providers and adjustable inference parameters like temperature.

- Through the Amazon Bedrock console, you can access additional capabilities such as safeguards, orchestration, model assessment, and deployment management.

You can use a single **Amazon Bedrock API** to securely access FMs. By using the same API, you can privately pass prompts and responses between the user and the FM.

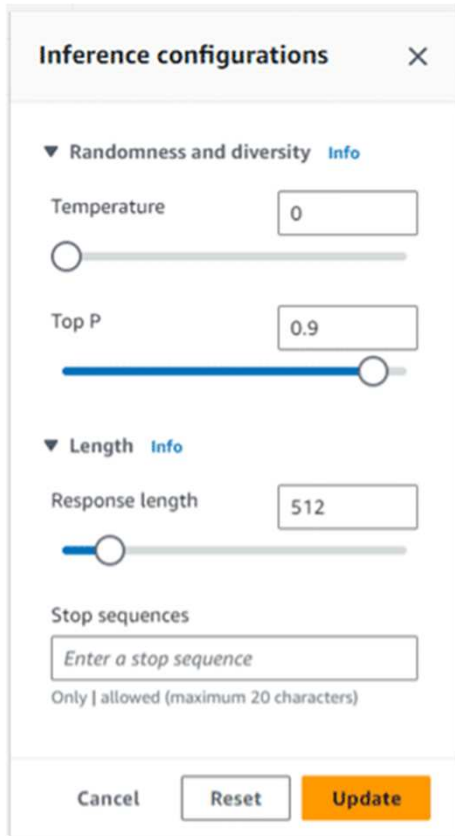
How do you interact with the Amazon Bedrock playground?

You can access Amazon Bedrock with the AWS Management Console to use the text, chat, or image playground.



Consumers interact with Amazon Bedrock in the console for chat, text, and image playground activity with the FMs.

How do you interact with the Amazon Bedrock playground?



The screenshot shows the 'Inference configurations' dialog box in the Amazon Bedrock playground. It has a title bar with a close button (X). The dialog is divided into two main sections: 'Randomness and diversity' and 'Length'. The 'Randomness and diversity' section includes a 'Temperature' slider set to 0 and a 'Top P' slider set to 0.9. The 'Length' section includes a 'Response length' slider set to 512 and a 'Stop sequences' text input field with the placeholder 'Enter a stop sequence'. Below the text input is a note: 'Only | allowed (maximum 20 characters)'. At the bottom of the dialog are three buttons: 'Cancel', 'Reset', and 'Update'.

Inference configurations

▼ Randomness and diversity Info

Temperature 0

Top P 0.9

▼ Length Info

Response length 512

Stop sequences

Enter a stop sequence

Only | allowed (maximum 20 characters)

Cancel Reset Update

You can optionally set inference parameters to influence the response generated by the model. FMs support the following types of inference parameters.

- Temperature controls the randomness of a language model's output – lower values make responses more focused and predictable, while higher values allow for more diverse and creative word choices.
- Top P controls the diversity of a model's output by selecting from the smallest set of tokens whose cumulative probability exceeds the set threshold – higher values increase randomness, while lower values improve predictability.
- Response Length maximum number of tokens to use in a response
- Stop sequences

How do you interact with the Amazon Bedrock playground?

Stop sequences

- You can define specific character sequences—called **stop sequences**—that act as a signal to **end the model's output** early.

For example:

If you're generating text and you set a stop sequence like "###", then:

- As soon as the model outputs "###", it **immediately stops** generating any further text—even if the maximum token limit hasn't been reached.

Knowledge Check

Q1.

Which problems does Amazon Bedrock solve? (Select THREE.)

- A. Customers want access to multiple models so they can choose the model that best fits their needs.
- B. Customers want the models fine-tuned with their data to be private.
- C. Customers want to use foundation models (FMs) on premises.
- D. Customers do not want to manage their infrastructure.
- E. Customers want to build and train foundation models (FMs) from scratch.
- F. Customers want a coding companion.

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Q2.

What are the benefits of using Amazon Bedrock? (Select THREE.)

- A. Prompts and responses are not shared with AWS or third-party providers.
- B. Amazon Bedrock provides separate APIs to access different foundation models (FMs).
- C. Amazon Bedrock makes it possible to customize foundation models (FMs) privately.
- D. Amazon Bedrock provides access to a wide variety of foundation models (FMs) from Amazon and third-party providers.
- E. Amazon Bedrock requires customers to provision their own infrastructure.
- F. Amazon Bedrock trains third-party models with a customer's prompts and responses.

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Which option is a common use case for Amazon Bedrock?

- A. Demand forecasting
- B. Conversational AI assistants with relevant data
- C. Extraction of text from images
- D. Predictive maintenance of vehicles

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